

Data Visualization

Objective

The objective of the Data Visualization phase is to transform the prepared agricultural dataset into meaningful and interactive visualizations using Tableau. These visualizations help users understand crop production trends, seasonal variations, regional productivity, and overall agricultural performance across India.

Description

After completing data preparation, the cleaned dataset was imported into Tableau for visualization. Various charts and interactive dashboards were created to analyze India's agricultural crop production from **1997 to 2021**.

A total of **9 unique visualizations** were developed, including KPI cards, line charts, bar charts, an India map, pie charts, and donut charts. These visualizations allow users to compare crop production across different states, identify seasonal trends, evaluate agricultural productivity, and explore long-term production patterns. The interactive dashboards make it easier for farmers, policymakers, and investors to derive valuable insights and make informed decisions.

Visualizations Created

- KPI Cards
- Line Chart
- India Map
- Horizontal Bar Chart
- Grouped Bar Chart
- Donut Chart
- Pie Chart
- Seasonal Crop Analysis
- Crop Production Comparison

Key Outcomes

- Successfully created 9 interactive Tableau visualizations.
- Enabled state-wise and seasonal crop production analysis.
- Identified production trends from 1997–2021.
- Developed dashboards for effective decision-making.
- Improved data interpretation through interactive visual analytics.

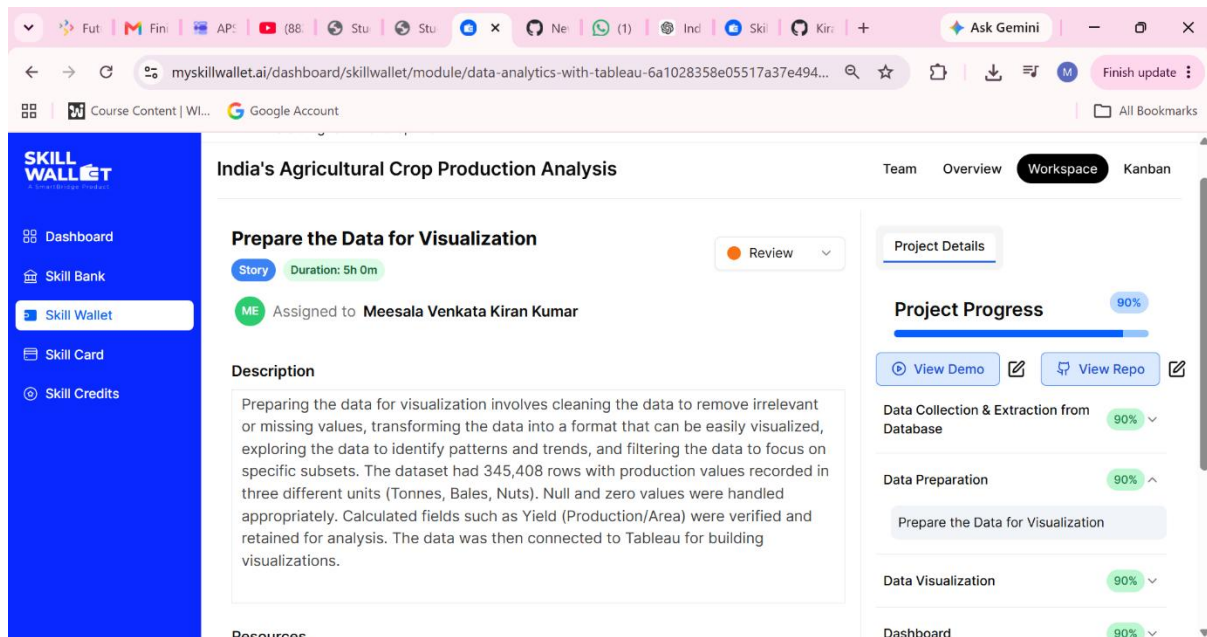


Figure 1: Prepare the Data for Visualization

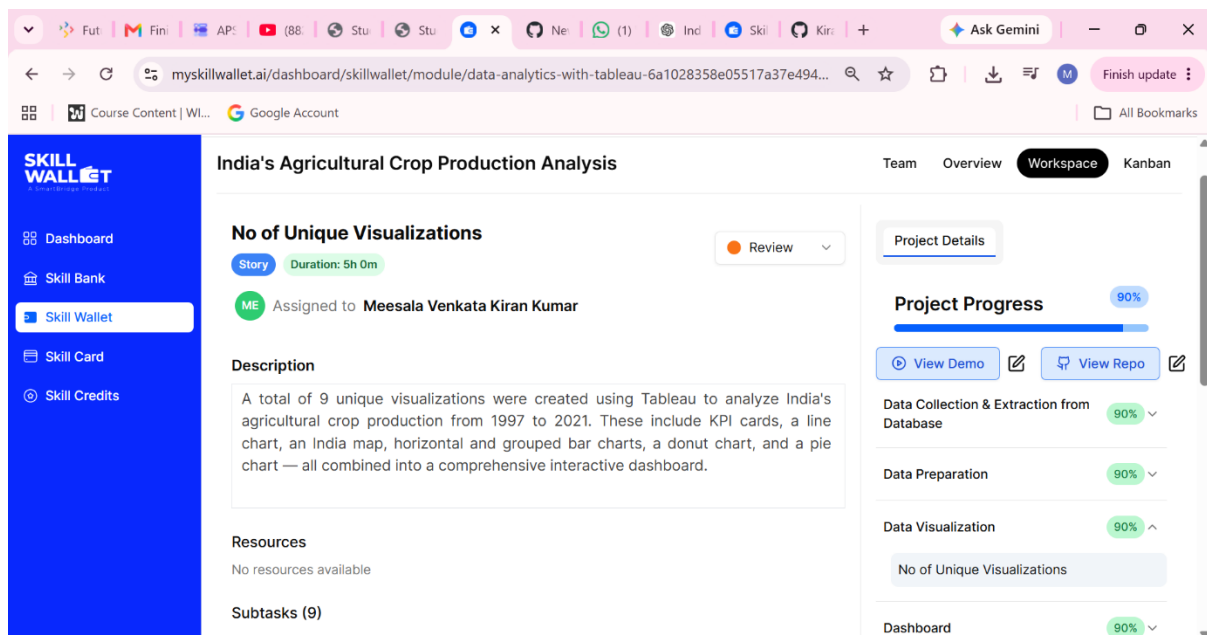


Figure 2: Number of Unique Visualizations

Conclusion

The Data Visualization phase successfully converted the prepared agricultural dataset into interactive and insightful Tableau visualizations. These dashboards provide a comprehensive understanding of crop production, seasonal trends, and regional performance, enabling stakeholders to make informed, data-driven decisions for agricultural planning and development.